Here is the fully organized and hardened interactive learning environment kit based on your high-dimensional spatial relationships documentation and specifications.

The files below resolve all markup issues, add robust error handling for zero-vectors and dimension mismatches, incorporate MathJax for LaTeX formatting, and upgrade the input mechanisms to textarea inputs for seamless scaling to 768D, 1536D, and higher-dimensional embeddings.

1. interactive_glossary.html

Save this file as interactive_glossary.html. It incorporates responsive layout structures, dark mode support, and the interactive calculator modules.

HTML

```
<!DOCTYPE html>

 html lang  "en"

 head

    meta charset  "UTF-8"

    meta name  "viewport" content  "width=device-width, initial-scale=1.0"

    title                                                    title


  <!-- MathJax Configuration -->

    script

     window

        tex

            inlineMath    '$'  '$'    '\\('  '\\)'

            displayMath    '$$'  '$$'    '\\['  '\\]'



     script

    script async src  "https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-mml-chtml.js"     script
```

```
  <!-- Stylesheet -->
    link rel  "stylesheet" href  "styles.css"
 head
body data-theme  "light"

div class  "app"
  <!-- Sidebar Navigation -->
    aside class  "sidebar"
       h2                    h2
       button class  "button" onclick  "toggleTheme()" style  "width:100%; margin-bottom: 20px;" 🌙
☀️            button
       div class  "sidebar-nav"
          a href  "#foundations"                  a
          a href  "#metrics"                   a
          a href  "#umap-section"                      a
          a href  "#hyperbolic"                       a
          a href  "#calculator-section"                a
        div
     aside

  <!-- Main Content Area -->
    main class  "main"
```

```
        h1                                                                  h1

        p class  "subtitle"
                                    p


        div class  "controls"

            input id  "searchBox" class  "search-box" placeholder  "Search terms, metrics, or
equations..." onkeyup  "filterGlossary()"

         div


        div id  "glossary"

          <!-- Point -->

           div class  "card glossary-entry" data-name  "point coordinate location"

              div class  "category"                    div

              div class  "term" onclick  "toggle(this)"          div

              div class  "content"

                 p    strong                 strong
                                               p

                 div class  "visual"                      div

               div

            div


          <!-- Vector -->

           div class  "card glossary-entry" data-name  "vector direction length"

              div class  "category"                    div
```

```
    div class  "term" onclick  "toggle(this)"              div
    div class  "content"
      p
                                                                                  p
      div class  "math"                                                  div
    div
  div


<!-- Input vs Output Metric -->
  div class  "card glossary-entry" data-name  "umap input metric output metric geometry"
    div class  "category"              div
    div class  "term" onclick  "toggle(this)"                                  div
    div class  "content"
      p    strong                              strong
                 p
      p    strong                                  strong
                 p
    div
  div


<!-- Hyperbolic UMAP -->
  div class  "card glossary-entry" data-name  "hyperbolic hyperboloid poincare disk umap"
    div class  "category"                          div
```

div class "term" onclick "toggle(this)"
div

div class "content"

p

strong strong

p

div class "formula"

$$DiskProjection: X = \frac{x}{1+z}, Y = \frac{y}{1+z}$$

div

div

div

div

<!-- Interactive Advanced Mathematics Lab -->

div class "card" id "calculator-section"

div class "category" div

div class "term" onclick "toggle(this)"
div

div class "content"

p

p

label strong strong label

```
            textarea id  "vectorA" rows  "3"              textarea


            label    strong                  strong     label

            textarea id  "vectorB" rows  "3"              textarea


            button class  "button" onclick  "calculateMetrics()" style  "margin-top:10px;"
          button


            div id  "results"     div

          div

        div

      main

    div


  script src  "app.js"     script

  body

  html
```

2. styles.css

Save this file as styles.css. It provides modern layout rules, responsive grid setups, and dark/light token management.

CSS

```
:root

        #f5f7fb

            #ffffff
```

#eef4ff

#1b1b1b

#666

#3366ff

#2148c0

#0a8f47

#d11f1f

#d6dce5

14px

0 4px 12px rgba 0 0 0 08

[data-theme="dark"]

#111827

#1f2937

#293549

#f4f4f4

#c8c8c8

#344255

body

0

var

var

"Segoe UI"

1.6

.app

280px 1

100vh

.sidebar

var

1px var

20px

0

100vh

.sidebar a

var

10px

8px

4px

.sidebar a:hover

var

.main

30px

1100px

.card

var

1px var

var

var

22px

20px

.term

20px

var

.category

var

11px

700

0.5px

6px

.content

12px

.search-box

100%

14px

16px

1px var

10px

var

var

20px

.button

var

12px 18px

10px

600

.button:hover

var

textarea

100%

12px

10px

1px var

var

var

.formula .math

12px

12px 0

var

8px

.visual

#0d1117

#00ff95

8px

12px

10px

.result-box

20px

16px

10px

var

5px var

@media 900px

.app 1

.sidebar



3. app.js

Save this file as app.js. It includes single-pass metric math optimization, comprehensive validation against empty arrays, dimension mismatches, near-zero vector protection, and automatic MathJax typesetting triggers.

JavaScript

```
function toggleTheme

    const            document

                                                    "dark"    "light"   "dark"




function filterGlossary

    const            document                    "searchBox"

    document                        ".glossary-entry"

        let

                                                  "block"   "none"




function toggle

    const

                                                      "none"     "block"   "none"




function parseVector

    const                                /[\s,]+/                    ""          Number
```

```
    if                  isNaN
        throw new Error "Vectors contain non-numeric tokens or formatting errors."

    return new Float64Array


function calculateMetrics
    const               document            "results"
    try
        const           document            "vectorA"
        const           document            "vectorB"

        if              0               0
            throw new Error "Vectors cannot be empty."

        if
            throw new Error `Dimension mismatch: Vector A has ${A.length} dimensions while Vector B has ${B.length} dimensions.`


        const
        let     0
        let             0
```

```
let                0

let                   0

let              0


// Single-pass O(n) metric calculation loop

for  let     0

    const

    const

    const


                        Math


const             Math

const             Math

const                 Math


const          1e-12

let            null
```

```
let                 null

if

            Math        1  Math       1              // Clamp floating point bounds

              Math                    180   Math



const                                 null    "Undefined (Zero-vector detected)"                  8

const                                  null    "Undefined"                      4     "°"

                                `

   <div class="result-box">

      <h3>Calculation Results (${n} Dimensions)</h3>

      <p><b>Euclidean Distance:</b> ${euclidean.toFixed(6)}</p>

      <p><b>Manhattan Distance:</b> ${manhattan.toFixed(6)}</p>

      <p><b>Dot Product:</b> ${dot.toFixed(6)}</p>

      <p><b>Vector A Norm:</b> ${normA.toFixed(6)}</p>

      <p><b>Vector B Norm:</b> ${normB.toFixed(6)}</p>

      <p><b>Cosine Similarity:</b> ${cosineDisplay}</p>

      <p><b>Angle Between Vectors:</b> ${angleDisplay}</p>

   </div>

`
```

```
catch
                                `
    <div class="result-box" style="background:#ffe6e6; border-left-color: var(--danger);">
        <h3 style="color: var(--danger);">Validation Error</h3>
        <p>${err.message}</p>
    </div>
`
```

